Dyuti Ballav Paul

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Education

Indian Institute of Technology (Indian School of Mines) Dhanbad October 2022 - May 2026

Bachelor of Technology in Computer Science and Engineering

- *CGPA: 8.91 / 10.00*
- **Relevant Coursework:** Data Structures, Algorithm Design and Analysis, Operating Systems, Object-Oriented Programming, Database Management Systems, Computer Networks, Software Engineering, Probability & Statistics, Linear Algebra, Optimization Techniques, IoT, Information Retrieval, Advanced Algorithms, TinyML and Efficient AI Computing

Experience

Samsung R&D Institute May 2025 – July 2025

Noida, Uttar Pradesh *Internship*

- Enhanced the **Samsung live translation** during calls and **Bixby Text Call** by preserving emotion features through the Speech-to-Text stage of the 3-stage pipeline for downstream emotion detection.
- Pretrained audio-classification models are fine-tuned on open-source emotionally labeled datasets; used **VITS** for **TTS** with emotion-mapped speakers, merging LJSpeech with an emotion TTS dataset for broader vocabulary coverage.
- Converted models to **ONNX** and applied quantization for **on-device** deployment — reduced TTS model size by $\sim$**75%** (i.e., size $\sim$**100 MB**), achieving $\sim$**2 s** in emotion detection and $\sim$**1 s** TTS latency.

Samsung R&D Institute August 2026 – Present

Noida, Uttar Pradesh

Optimizing edge AI capabilities for resource-constrained hardware and mobile platforms.

Projects

Smart Traffic Control System (STC) | *PyTorch, FastAPI, React.js, Eclipse SUMO, MQTT, InfluxDB, Unity 3D* GitHub

- Built an adaptive multi-intersection traffic signal controller using deep RL on a real-world road network (Patna STC) simulated via ***Eclipse SUMO Digital Twin***.
- *Architecture* – Designed a Spatial-Temporal Graph Attention Network (STGAT) with 3-head ***Graph Attention***, $K=3$ ***hop residual stacking***, ***GRU*** temporal memory, BFS-based graph adjacency, and demand-responsive ***action masking*** for minimum green-time safety constraints.
- *Backend* – Deployed ***FastAPI*** with ***WebSockets*** for sub-second inference streaming to clients.
  - Integrated ***Mosquitto MQTT*** broker for Fog Computing edge sensor telemetry aggregation.
  - Configured ***InfluxDB*** for time-series persistence. Orchestrated services via ***Docker Compose***.
- *Frontend* – Developed dashboard using React.js (Vite), ***Recharts*** for live metrics, and ***Unity 3D*** for 3D spatial rendering.

Smart Hospital | *TypeScript, React.js, Tailwind CSS, Node.js, Express.js, Postgres, Firebase* GitHub

- Built a hospital portal for appointment booking, doctor prescriptions, patient prescription tracking, and admin staff management.
- *Backend* – Node.js, Express.js with ***PostgreSQL*** and ***Prisma ORM***.
  - *Email or phone* for authentication and authorization. **Firebase** used for sending OTP to the phone.
  - ***Razorpay*** payment gateway integrated with ***Webhooks*** for secure payment confirmation.
- *Frontend* – TypeScript, React.js, Tailwind CSS, Shadcn

Blogging app | *TypeScript, React.js, Hono framework, Postgres, Prisma ORM, Cloudflare Workers* GitHub

- Built a blogging platform where users can create, read, and manage posts.
- *Backend* – Serverless backend using ***Hono*** framework on ***Cloudflare Workers***.
  - Used ***Prisma Accelerate*** for ***connection pooling*** to manage the number of connections to the database.
  - Implemented ***JWT***-based authentication.
- *Frontend* – Used TypeScript, React.js, Tailwind CSS and Vite as a bundler. Frontend is deployed in ***Vercel***.

Technical Skills

Languages: C, C++, Java, Python, JavaScript, TypeScript
Frontend: HTML, CSS, React.js, Next.js, Tailwind CSS
Backend: Spring Boot, Node.js, Express.js, REST APIs, Hono framework, JWT
Databases and ORM/ODM: PostgreSQL, MySQL, MongoDB, Prisma ORM, Mongoose
AI & ML: PyTorch, TensorFlow, Hugging Face.
Tools and DevOps: Git, GitHub, Postman, Docker, CI/CD, AWS, Cloudflare Workers